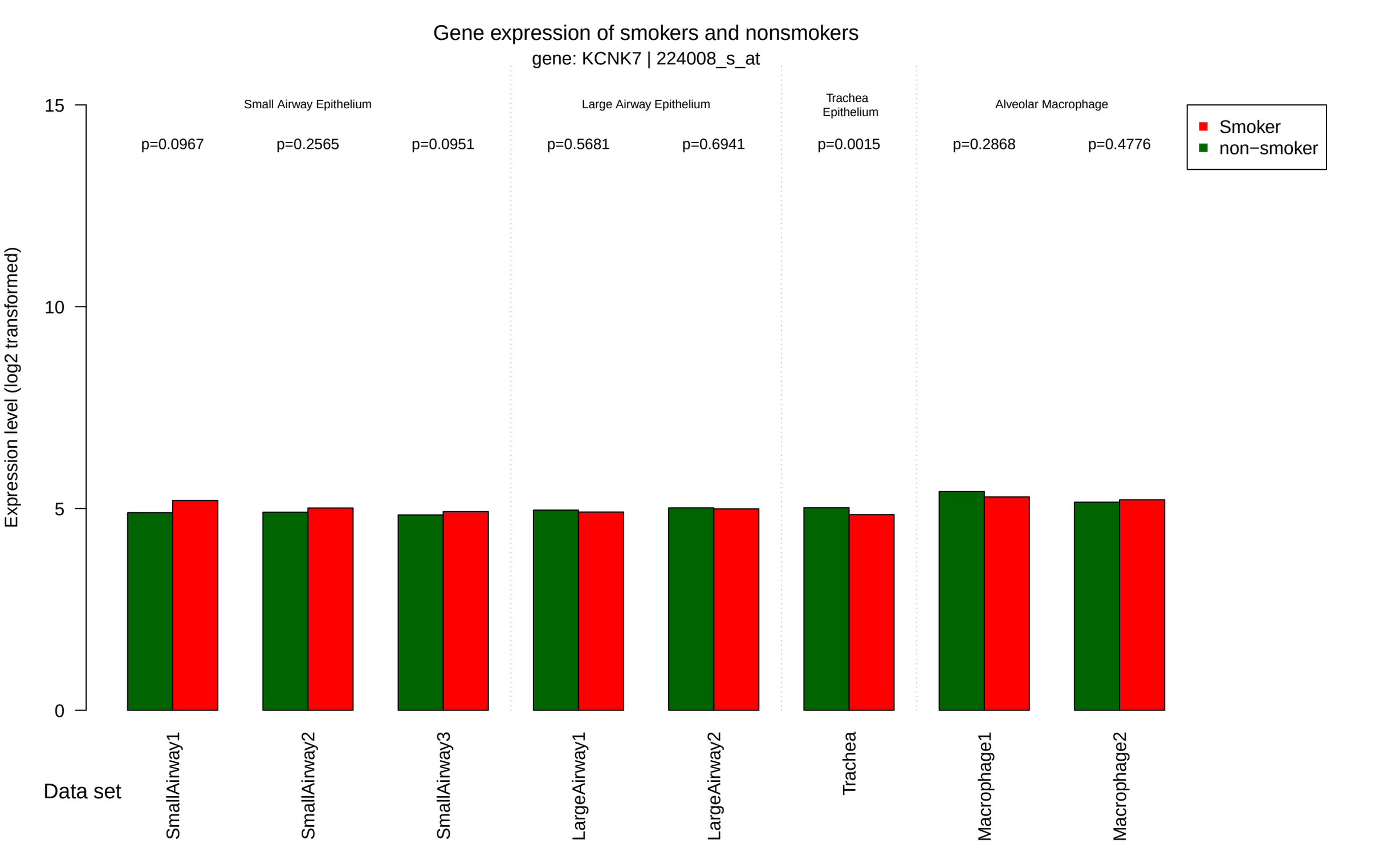
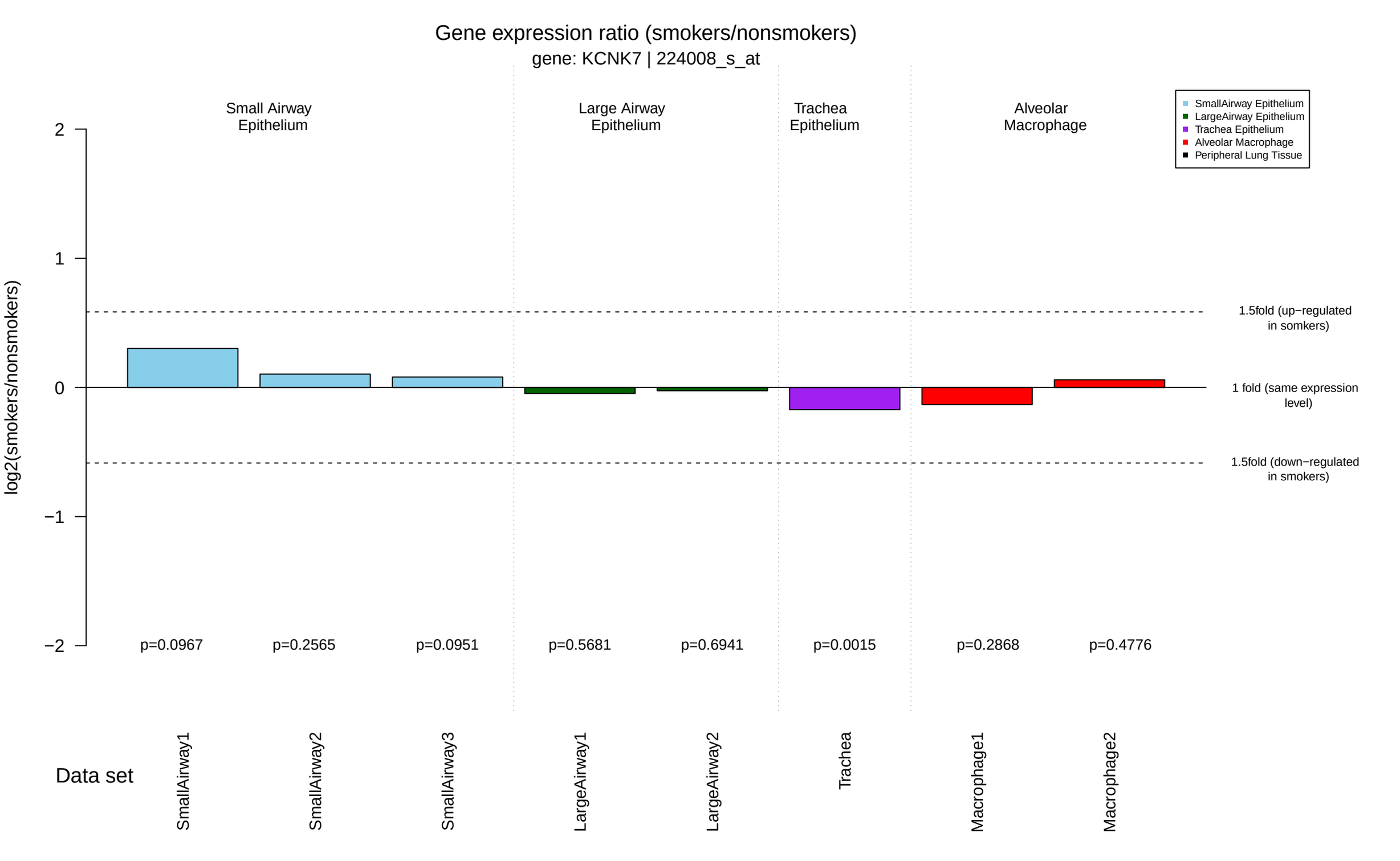
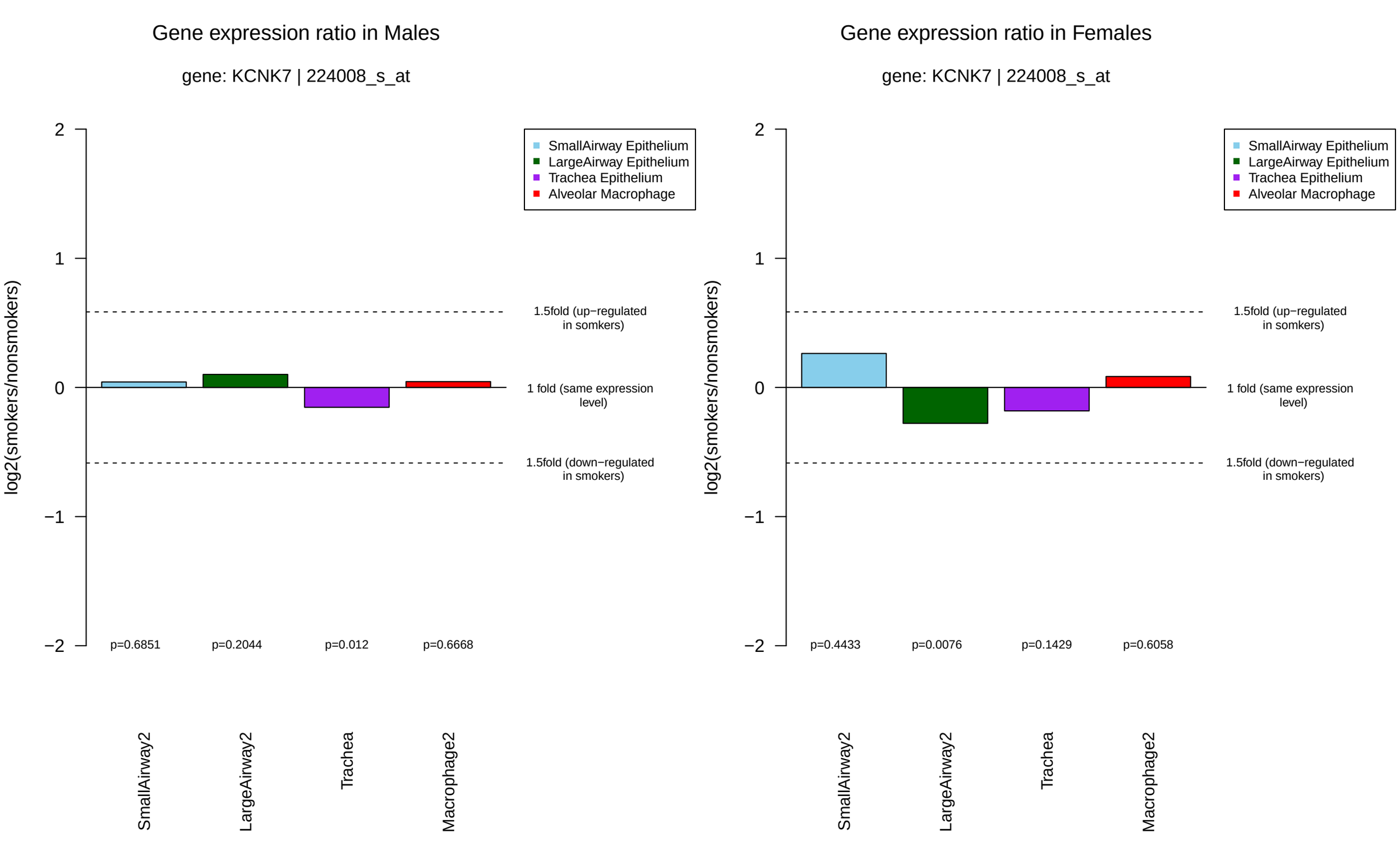
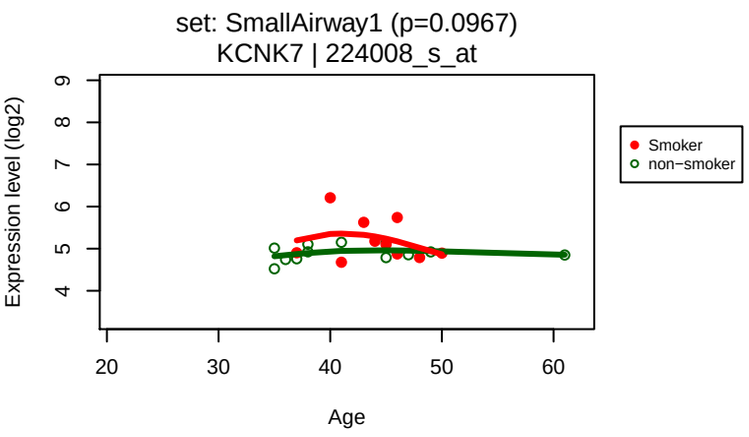
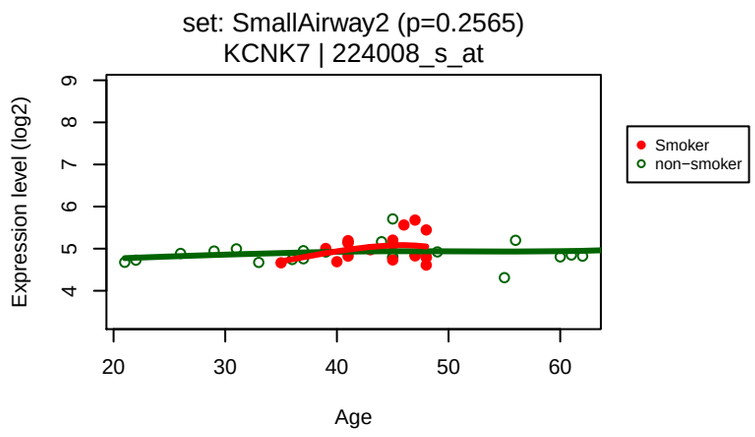
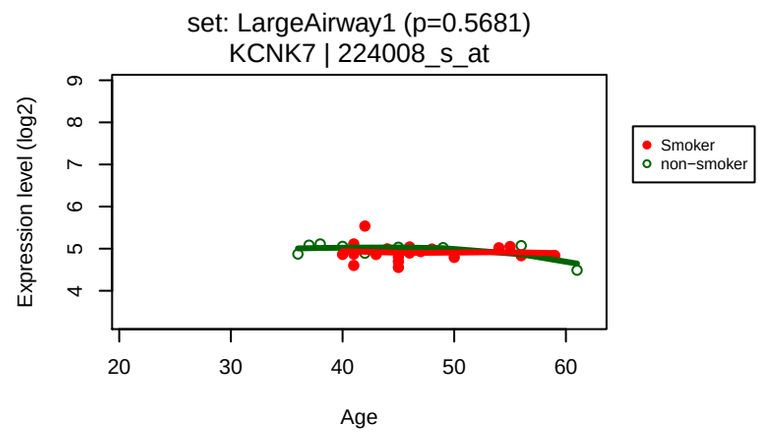

Gene expression of smokers and nonsmokers
gene: KCNK7 | 224008_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0967
p=0.2565
p=0.0951
p=0.5681
p=0.6941
p=0.0015
p=0.2868
p=0.4776
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

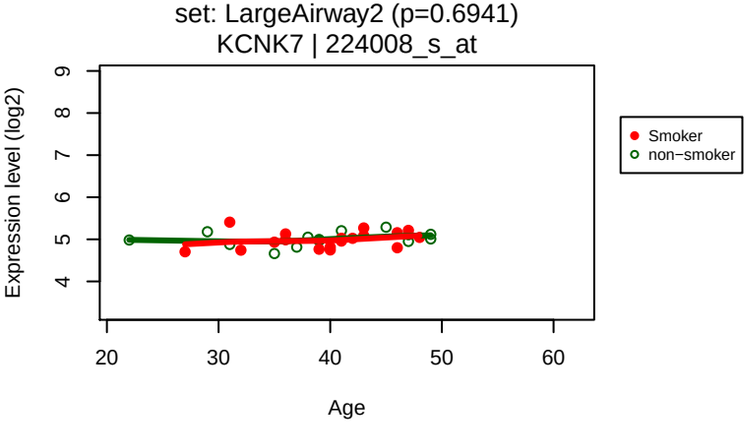

Gene expression ratio (smokers/nonsmokers)
gene: KCNK7 | 224008_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0967
p=0.2565
p=0.0951
p=0.5681
p=0.6941
p=0.0015
p=0.2868
p=0.4776
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

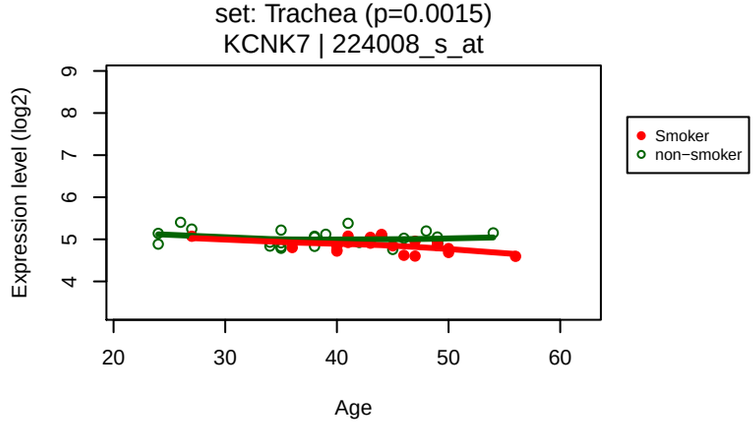

Gene expression ratio in Males
gene: KCNK7 | 224008_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6851
p=0.2044
p=0.012
p=0.6668
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KCNK7 | 224008_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4433
p=0.0076
p=0.1429
p=0.6058
SmallAirway2
LargeAirway2
Trachea
Macrophage2

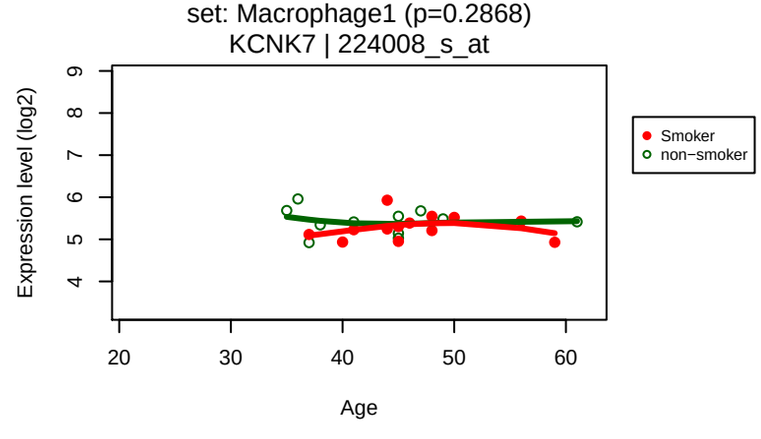

set: SmallAirway1 (p=0.0967)
KCNK7 | 224008_s_at
Expression level (log2)
Age
Smoker
non−smoker

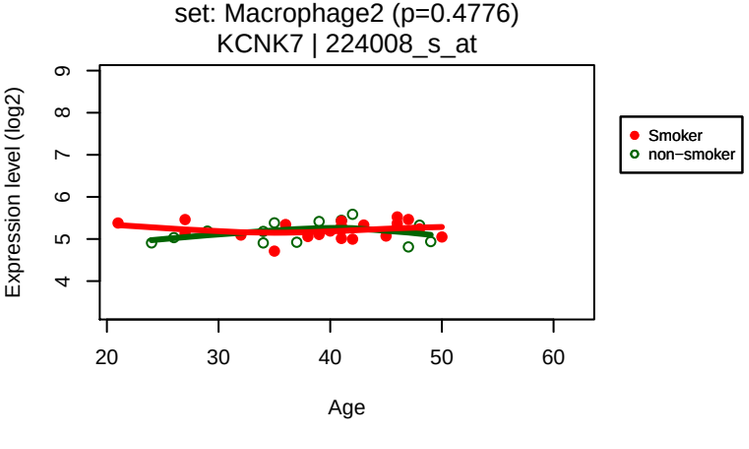

set: SmallAirway2 (p=0.2565)
KCNK7 | 224008_s_at
Expression level (log2)
Age
Smoker
non−smoker

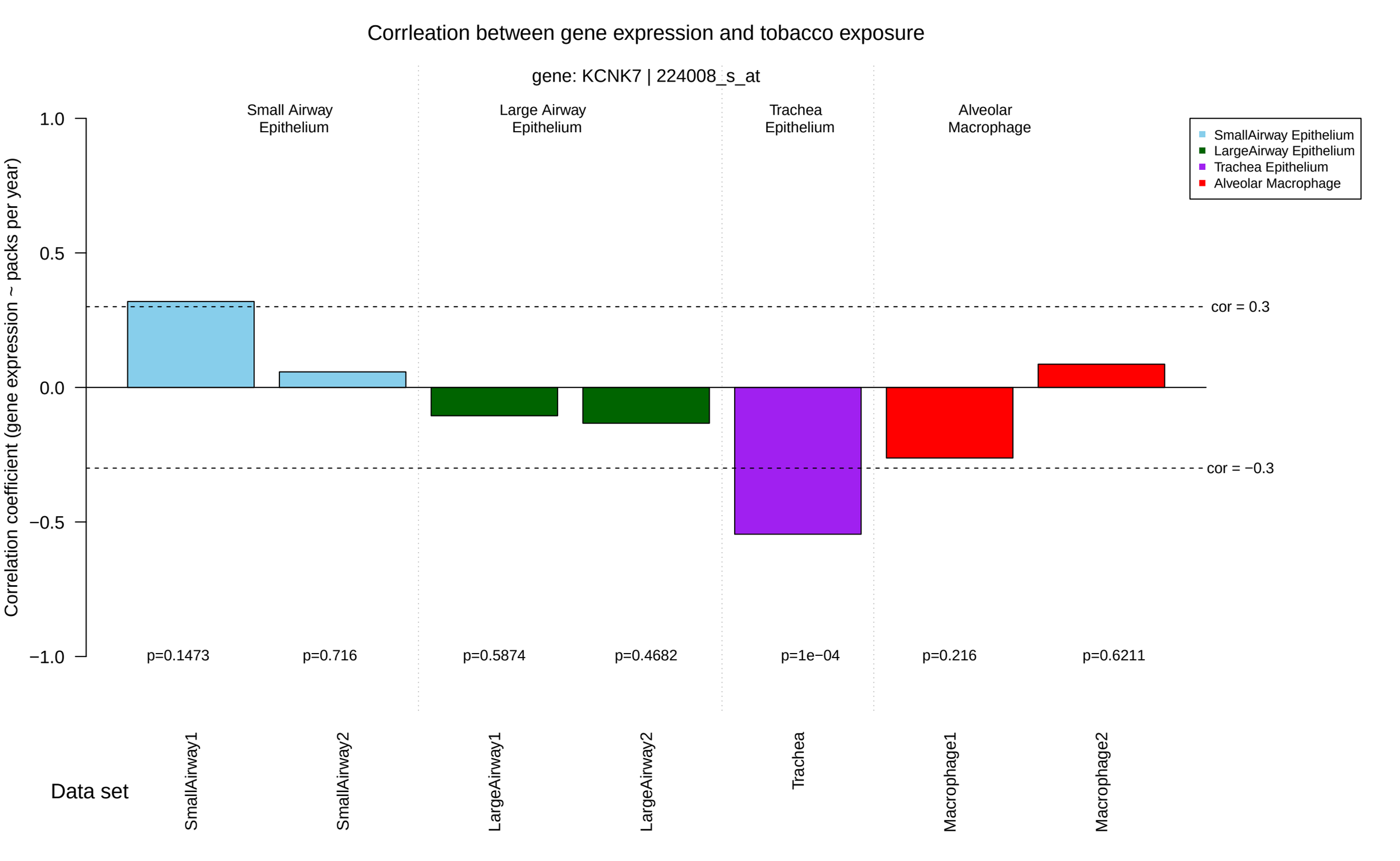

set: LargeAirway1 (p=0.5681)
KCNK7 | 224008_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6941)
KCNK7 | 224008_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0015)
KCNK7 | 224008_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2868)
KCNK7 | 224008_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4776)
KCNK7 | 224008_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KCNK7 | 224008_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1473
p=0.716
p=0.5874
p=0.4682
p=1e−04
p=0.216
p=0.6211
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2